

LEICESTER, UK

JAMES MASON

PRODUCT DESIGN ENGINEER — AUDIO, PROTOTYPING & CREATIVE TECHNOLOGY

Product design engineer in audio since 2017. I take loudspeakers, hi-fi products and electronics enclosures from brief through CAD, drawings and prototypes to manufacture, working with electronics and acoustic engineers, fabricators, suppliers and the workshop. Alongside that I design generative music tools, setting the concept, rules and test criteria, with the software built using AI coding tools.

EXPERIENCE

DESIGN ENGINEER, TPI AEROSPACE LIMITED

JAN 2019 – PRESENT

Loudspeakers, hi-fi, PA and mixer hardware from brief to production: SharkCAD models and drawings, BOM systems in Google Sheets, supplier specifications and prototyping with fabricators.

- Hired, with a CNC woodwork programmer and operator, to bring TPI's loudspeaker cabinet manufacture in-house: re-engineered the existing designs, including TOTEM (now TOTEM 2) and MFR, to suit the workshop, solving recurring assembly issues, improving quality consistency and reducing build cost and time
- Design and engineering of the Type. HF hi-fi range and of the cinema and Movement ranges, through to production drawings
- The only product designer on Clarity A and Clarity M for Mastersounds
- Mechanical design of the modular Type. 29 analogue mixer: chassis, panels and module system
- Spotted the need for, and developed, a 3D crate-packing tool now used to plan TPI's shipping crates (software written with AI coding tools I directed)

DESIGN ENGINEER, WILSON BENESCH

OCT 2017 – DEC 2018

Eminence and Precision loudspeakers and GMT One turntable development: SolidWorks models, drawings, BOMs, KeyShot renders and an in-house 3D-printing workflow.

- Helped develop the Eminence flagship loudspeaker; designed its Fibonacci-pattern dust caps and decoupled tweeter faceplate with a PhD student on placement
- Developed the Precision Series entry-level range: CAD, technical drawings and marketing renders
- Drawings, models and manufacturing files for the initial GMT One turntable prototype
- Researched, specified and set up in-house 3D printing

EDUCATION

BA (Hons) Product Design, First Class – Sheffield Hallam University, 2014–2017

SKILLS

DESIGN AND CAD

SharkCAD, SolidWorks, KeyShot, Blender, Affinity Designer

MAKING

Sheet metal, solid surface, carbon fibre (VRTM), 3D printing, assembly and finishing

SYSTEMS

Google Sheets BOMs and calculators, production documentation

CREATIVE TECHNOLOGY

Generative music systems, interaction and parameter design, listening evaluation, AI-assisted development

EARLIER EXPERIENCE

BiesseWorks, Mazatrol, MRPeasy, Adobe Photoshop, Illustrator and InDesign

ALSO

Full UK driving licence.

CONTACT

jamesjmason3@gmail.com jamesmasondesign.com

JAMES MASON • PRODUCT DESIGN ENGINEER • LEICESTER, UK